Project Proposal

TIH - IoT CHANAKYA Fellowship Program 2026

Image-based Disease Detection leveraging Machine Learning Approaches for Turmeric (Curcuma longa) and Ridge Gourd (Luffa acutangula)

Submitted by

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July 11, 2026

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CIN: U73200MH2020NPL344467

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Version 1.0
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Executive Summary

Plant diseases account for an estimated 20–40% of global crop production losses annually, and 15–25% of yield losses in India, translating to roughly Rs. 0.9–1.4 lakh crore in economic damage each year. Manual diagnosis by farmers or extension officers remains slow, subjective and reactive, delaying intervention until the cost-effective treatment window has often closed. This project addresses this gap by building, comparing, and deploying image-based disease detection models for two agronomically important yet under-studied crops: Turmeric (Curcuma longa) and Ridge Gourd (Luffa acutangula), selected for their contrasting disease etiology — turmeric's dominant threat is fungal/rhizome disease expressed through subtle foliar symptoms, while ridge gourd is dominated by viral Yellow Mosaic Disease (ToLCNDV) causing distinct mottling and leaf distortion. Using the labelled dataset provided by TIH-IoT, we will train and rigorously benchmark four to five candidate architectures per crop — spanning lightweight CNNs (MobileNetV3), mid-size CNNs (EfficientNet-B0), deep CNNs (ResNet-50), Vision Transformers (ViT-B/16), and a hybrid CNN-ViT model — evaluated jointly on accuracy, field robustness, computational cost, calibration, and interpretability. The best-performing model per crop will be compressed through distillation and INT8 quantization and deployed via a hybrid edge-and-cloud architecture combining on-device offline inference with a cloud-hosted API, supported by Grad-CAM based explainability and a documented retraining loop for continuous improvement. Deliverables include benchmarked models, a deployment-ready inference pipeline, field-validation reports, and complete technical documentation, over a milestone-driven, 10-month project plan.

1 Background

Crop disease has historically been one of the most persistent threats to agricultural productivity worldwide; pest-and-disease losses in India rose from roughly 7% of output in the early 1960s to over 23% by the early 2000s, driven by intensified cultivation and the limited reach of expert plant pathology services. In India, where agriculture contributes nearly 28% of GDP, disease diagnosis has traditionally relied on visual inspection by farmers or periodic extension-officer visits — a model that has not scaled with the pace or geographic spread of outbreaks. This project targets two crops with contrasting disease profiles. Turmeric, India's major GI-tagged export spice, suffers primarily from rhizome rot and leaf blotch, fungal diseases that manifest as visible but subtle foliar symptoms well before the economically damaging rhizome infection is confirmed. Ridge gourd, a widely cultivated smallholder vegetable, is increasingly threatened by Yellow Mosaic Disease caused by the whitefly-transmitted begomovirus ToLCNDV, producing leaf mottling, chlorosis, and distortion that is spreading across Indian growing regions with almost no existing image-based diagnostic support.

2 Statement of the problem

Farmers cultivating Turmeric (Curcuma longa) and Ridge Gourd (Luffa acutangula) currently lack a fast, reliable, scalable means of diagnosing foliar disease at the point of observation, resulting in delayed intervention and avoidable yield loss. Existing computer-vision research concentrates on a small set of heavily benchmarked crops such as tomato and potato; turmeric and ridge gourd remain comparatively under-studied, with essentially no published image-based diagnostic work for ridge gourd's ToLCNDV-driven Yellow Mosaic Disease. This project addresses that gap by developing, comparing, and deploying image-based deep learning models for accurate, field-robust disease detection in both crops.

3 Objectives

Project objectives (SMART: Specific, Measurable, Action-oriented, Realistic, Time-based):

(1) Curate and validate a versioned, leakage-free image dataset for Turmeric and Ridge Gourd from TIH-IoT's labelled data within Month 1; (2) train and benchmark five candidate architectures (MobileNetV3, EfficientNet-B0, ResNet-50, ViT-B/16, hybrid CNN-ViT) per crop against defined accuracy, robustness and latency targets by Month 5; (3) optimize the best model via compression and quantization for edge deployment, capping field-accuracy loss at 2 points, by Month 7; (4) deploy a monitored, explainable pipeline validated through two field-trial rounds by Month 9; and (5) deliver complete documentation and SOPs by Month 10.

4 Deliverables

- The project delivers a compared, optimized, deployment-ready disease detection system for Turmeric and Ridge Gourd.
- Key deliverables: a versioned dataset and preprocessing pipeline; five benchmarked candidate models per crop (MobileNetV3, EfficientNet-B0, ResNet-50, ViT-B/16, hybrid CNN-ViT); one compressed, quantized deployment model per crop with Grad-CAM explainability; a hybrid edge-and-cloud pipeline (Android offline app, REST API, Docker); two field-validation reports; and complete documentation, SOPs, and a reproducible codebase.
- Quantitative targets: macro-F1 above 90% (lab) and 85% (field) per crop, under 2 points of field-accuracy loss post-quantization, and on-device inference latency under 500 ms.

5 Project Plan with Major Milestones

- All projects must be delivered within a specific timeframe as specified in the call..
- The major milestones to be achieved should be specified clearly in the project plan.
- Monthly milestones can be specified in the project plan.

Sr. No.	Milestone	Target Date	Remarks
1	Data ingestion, checksumming, schema validation and class-distribution audit for Turmeric and Ridge Gourd datasets	Month 1	Versioned dataset manifest v0 finalized with TIH-IoT sign-off
2	Preprocessing pipeline, augmentation policy and leakage-free stratified train/val/test split finalized	Month 2	inference_core module v1 packaged
3	Baseline models trained: traditional-ML classifier and MobileNetV3 (transfer learning)	Month 3	First model-leaderboard entries per crop
4	Deeper CNN and Transformer candidates trained: EfficientNet-B0 and ViT-B/16	Month 4	CNN vs. Transformer field-robustness comparison drafted
5	Full four-model comparison completed; Field-Validation Round 1 conducted with TIH-IoT	Month 5	Faculty/TIH-IoT go/no-go checkpoint on model selection

6	Hybrid CNN-ViT model trained (optimization phase); ablation study completed	Month 6	Novel architecture benchmarked against four baselines
7	Model compression via distillation and INT8 quantization; explainability (Grad-CAM) integrated	Month 7	Deployable ONNX artifact with accuracy-retention report
8	REST API, Docker containers and offline-capable desktop application built and staged	Month 8	Staging deployment complete, internally tested
9	Edge (Android) deployment, monitoring/drift-detection stack and Field-Validation Round 2	Month 9	Live monitoring dashboard and lab-vs-field accuracy comparison
10	Final validation, technical report, SOPs, reproducibility package and system handover to TIH-IoT	Month 10	Final report, demo and sign-off against CFP deliverables

6 Resources and budget

- The project requires a two-student core team (one CV/ML-focused, one with agricultural-science grounding) supervised by the faculty mentor, with periodic access to TIH-IoT domain experts for field validation.
- Equipment and materials: GPU compute for training and hyperparameter optimization (institute HPC/GPU cluster, with Colab Pro/Kaggle GPU tier as fallback); open-source ML tooling (PyTorch, ONNX Runtime, Optuna, MLflow, Docker); a mid-range Android reference device for on-device benchmarking; and cloud/IIT-B server hosting for the staging API in Months 8–9. No paid dataset acquisition is required, as labelled data is provided by TIH-IoT.
- This is only to get an idea about the project budget for the reviewers. TIH-IoT will not be providing any budget for the project development.

7 Appendix

- Brief about the background work accomplished by the faculty/ team of students in image analysis/ computer vision.

The proposing faculty mentor and student team bring relevant academic and infrastructural grounding to this project, though no prior work specific to agricultural image-based disease detection has yet been carried out. The CV/ML-focused student has coursework and lab-assignment experience in convolutional neural networks and image classification pipelines using PyTorch. The agricultural-science-grounded student contributes domain familiarity with crop pathology and field data collection practices relevant to Turmeric and Ridge Gourd cultivation. The faculty mentor's research group maintains ongoing work in applied machine learning and has access to institute-level GPU/HPC compute infrastructure that will be used for this project. This proposal represents the team's first dedicated research effort at the intersection of computer vision and agricultural disease diagnosis; no prior results in this specific domain are claimed.

- List of relevant publications of faculty, students

No directly relevant prior publications by the faculty mentor or student team currently exist in the area of agricultural image-based disease detection. Any publications arising from this project will be added to this list upon acceptance for peer review.